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Kaon and antikaon optical potentials in isospin asymmetric hyperonic matter

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Abstract. The medium modifications of the energies of kaons and antikaons in isospin asymmetric hyperonic matter are investigated using a chiral $SU(3)$ model. The isospin-dependent medium effects are important for asymmetric heavy-ion collision experiments, as well as relevant for the neutron star phenomenology as the bulk matter in the interior of the neutron star is highly isospin asymmetric. The effects of hyperons on the medium modifications of the kaons and antikaons in the strange hadronic matter are investigated in the present work and are seen to be appreciable for hadronic matter with large strangeness fractions. The study of the K-mesons in the asymmetric strange hadronic matter can be especially relevant for the compressed strange baryonic matter which can result from asymmetric heavy-ion collision experiments in the future accelerator facility FAIR at GSI.

PACS. 24.10.Cn Many-body theory – 13.75.Jz Kaon-baryon interactions – 25.75.-q Relativistic heavy-ion collisions

1 Introduction

The topic of study of the in-medium properties of hadrons is an important problem in strong-interaction physics, which has relevance for the high-energy heavy-ion collision experiments, as well as for neutron star phenomenology. In heavy-ion collision experiments, the medium modifications of the hadrons can be seen in different observables, like particle yield, particle spectra as well as in their collective flow. The study of medium modifications of K-mesons were initiated by Kaplan and Nelson [1], who suggested the possibility of antikaon condensation in the interior of the neutron stars due to the drop in the mass of the antikaons in the nuclear medium. However, recent experimental observations on neutron star phenomenology impose constraints on the nuclear equation of state (EOS). The EOS for the nuclear matter obtained using an effective model should be consistent with the astrophysical bounds to be acceptable as an EOS for neutron star matter [2,3]. Recently, the nuclear matter EOS has been investigated consistent with the neutron star phenomenology as well as data for collective flow in heavy-ion collision experiments [4]. The in-medium modification of kaon/antikaon properties can be observed experimentally [5–9] in relativistic nuclear collisions from their abundance, spectra as well as collective flow. There have also been intense theoretical investigations [10–19] on the kaon and antikaon properties in dense and hot hadronic matter, and to study the effects of such modifications on the observables of the high-energy heavy-ion collision experiments. There has been extensive research to obtain the in-medium properties of kaons and antikaons due to their relevance for the high-energy heavy-ion collision experiments. These have been calculated by using several methods like extracting the values by fitting to kaonic atom data, using chiral Lagrangians or by coupled-channel methods [20]. However, the kaon/antikaon potential in hot and dense hadronic matter remains still an unresolved issue.

Isospin effects in hot and dense hadronic matter [21] are important in isospin asymmetric heavy-ion collision experiments. Within the UrQMD model the density dependence of the symmetry potential has been studied by investigating observables like the π^-/π^+ ratio, the n/p ratio, the Δ^-/Δ^{++} ratio, NN scattering cross-section and pion flow for neutron-rich heavy-ion collisions [22]. Within the same approach, the isospin effects on the production of K^0 and K^+ [23] have also been studied.

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In the present investigation we use a chiral $SU(3)$ model [24,25] for the description of hadrons in the medium. The model has been used to study nuclear matter, finite nuclei, and hyperonic matter. The properties of the vector mesons have also been investigated in the nuclear medium within this model [25,26]. The energies of kaons (antikaons) in the (asymmetric) nuclear matter [27–29] were also studied within this framework and the isospin dependence was seen to be appreciable for high densities and hence will be particularly relevant in the future compressed baryonic matter (CBM) experiment at FAIR, GSI. The kaon and antikaon properties have been investigated and shown to be consistent with the low-energy KN scattering data [30,31]. In the present work, we investigate the effect of strangeness on the kaon and antikaon optical potentials in the isospin asymmetric hyperonic matter, consistent with the low-energy kaon nucleon scattering lengths for channels $I = 0$ and $I = 1$ [28,29].

The outline of the paper is as follows: in sect. 2 we shall briefly review the $SU(3)$ model used in the present investigation. Section 3 describes the medium modification of the $K(\bar{K})$-mesons in this effective chiral model. In sect. 4, we discuss the results obtained for the optical potentials of the kaons and antikaons and the isospin-dependent effects on these optical potentials in asymmetric hyperonic matter. Section 5 summarizes our results and discusses possible extensions of the calculations.

2 The hadronic chiral SU(3) × SU(3) model

The effective hadronic chiral Lagrangian used in the present work is given as

$$\mathcal{L} = \mathcal{L}_{kin} + \sum_{W=X,Y,V,\mathcal{A},u} \mathcal{L}_{BW} + \mathcal{L}_{vec} + \mathcal{L}_0 + \mathcal{L}_{SB}. \quad (1)$$

Equation (1) corresponds to a relativistic model of baryons and mesons adopting a nonlinear realization of chiral symmetry [32–34] and broken scale invariance (for details see [24–26]) as a description of the hadronic matter. The model was used successfully to describe nuclear matter, finite nuclei, hypernuclei and neutron stars. The Lagrangian contains the baryon octet, the spin-0 and spin-1 meson multiplets as the elementary degrees of freedom. In eq. (1), $\mathcal{L}_{kin}$ is the kinetic energy term, $\mathcal{L}_{BW}$ contains the baryon-meson interactions in which the baryon-spin-0 meson interaction terms generate the baryon masses. $\mathcal{L}_{vec}$ describes the dynamical mass generation of the vector mesons via couplings to the scalar fields and contains additionally quartic self-interactions of the vector fields. $\mathcal{L}_0$ contains the meson-meson interaction terms inducing the spontaneous breaking of chiral symmetry as well as a scale invariance breaking logarithmic potential. $\mathcal{L}_{SB}$ describes the explicit chiral symmetry breaking.

The baryon-scalar meson interactions generate the baryon masses and the parameters corresponding to these interactions are adjusted so as to obtain the baryon masses as their experimentally measured vacuum values. For the baryon-vector meson interaction terms, F-type (antisymmetric) and D-type (symmetric) couplings are possible. Here we will use the antisymmetric coupling [24,28] because, following the universality principle [35] and the vector meson dominance model, one can conclude that the symmetric coupling should be small. Additionally, we choose the parameters [24,28] so as to decouple the strange vector field $\phi_\mu \sim \bar{s}\gamma_\mu s$ from the nucleon (ideal mixing). A small deviation of the mixing angle from the ideal mixing [36–38] has not been taken into account in the present investigation.

The Lagrangian densities corresponding to the interaction for the vector meson, $\mathcal{L}_{vec}$, the meson-meson interaction $\mathcal{L}_0$ and the explicit chiral symmetry breaking term, $\mathcal{L}_{SB}$, have been described in detail in refs. [24,28].

To investigate the hadronic properties in the medium, we write the Lagrangian density within the chiral $SU(3)$ model in the mean-field approximation and determine the expectation values of the meson fields by minimizing the thermodynamical potential [25,26].

3 Kaon (antikaon) interactions in the chiral SU(3) model

In this section, we derive the dispersion relations for the $K(\bar{K})$ [39] and calculate their optical potentials in asymmetric [28,29] hadronic matter, taking into account the effects from hyperons. The modifications of the kaon and antikaon energies arise from the effects from scalar mesons, scalar-isovector δ-meson and due to interactions with the nucleons and hyperons. The interactions of kaons and antikaons with the baryon octet are generated by the vectorial Weinberg-Tomazawa term as well as by terms similar to the range term in chiral perturbation theory. It might be noted here that the interaction of the pseudoscalar mesons with the vector mesons, in addition to the direct pseudoscalar meson-nucleon vectorial interaction leads to a double counting in the linear realization of the chiral effective theory [40]. Within the nonlinear realization of the chiral effective theories, such an interaction does not arise in the leading or sub-leading order, but only as a higher-order contribution [40]. Hence the vector meson-pseudoscalar meson interaction will not be considered within the present investigation. In the following, we shall derive the dispersion relations for the kaons and antikaons and study the dependence of the kaon and antikaon optical potentials on the strangeness content of the matter and the isospin asymmetric parameter, $\eta = \frac{1}{2}(\rho_n - \rho_p)/\rho_B$. For this, we shall include the effects from isospin asymmetry originating from the scalar-isovector δ field, vectorial interaction with the nucleons, an isospin symmetric range term [28] as well as an isospin-dependent range term arising from the interaction with the nucleons [29]. In the present investigation, we include the effects from the hyperons to study the kaon and antikaon properties, which were not taken into account in the earlier works.

$$
\begin{aligned}
\mathcal{L}_K = & -\frac{i}{4f_K^2}\Big[\Big(2\bar{p}\gamma^\mu p + \bar{n}\gamma^\mu n - \bar{\Sigma^-}\gamma^\mu\Sigma^- + \bar{\Sigma^+}\gamma^\mu\Sigma^+ - 2\bar{\Xi^-}\gamma^\mu\Xi^- - \bar{\Xi^0}\gamma^\mu\Xi^0\Big) \\
& \times\Big(K^-(\partial_\mu K^+) - (\partial_\mu K^-)K^+\Big) \\
& +\Big(\bar{p}\gamma^\mu p + 2\bar{n}\gamma^\mu n + \bar{\Sigma^-}\gamma^\mu\Sigma^- - \bar{\Sigma^+}\gamma^\mu\Sigma^+ - \bar{\Xi^-}\gamma^\mu\Xi^- - 2\bar{\Xi^0}\gamma^\mu\Xi^0\Big) \\
& \times\Big(\bar{K^0}(\partial_\mu K^0) - (\partial_\mu \bar{K^0})K^0\Big)\Big] \\
& +\frac{m_K^2}{2f_K}\Big[(\sigma+\sqrt{2}\zeta+\delta)(K^+K^-) + (\sigma+\sqrt{2}\zeta-\delta)(K^0\bar{K^0})\Big] \\
& -\frac{1}{f_K}\Big[(\sigma+\sqrt{2}\zeta+\delta)(\partial_\mu K^+)(\partial^\mu K^-) + (\sigma+\sqrt{2}\zeta-\delta)(\partial_\mu K^0)(\partial^\mu \bar{K^0})\Big] \\
& +\frac{d_1}{2f_K^2}\Big(\bar{p}p + \bar{n}n + \bar{\Lambda^0}\Lambda^0 + \bar{\Sigma^+}\Sigma^+ + \bar{\Sigma^0}\Sigma^0 + \bar{\Sigma^-}\Sigma^- + \bar{\Xi^-}\Xi^- + \bar{\Xi^0}\Xi^0\Big) \\
& \times\big((\partial_\mu K^+)(\partial^\mu K^-) + (\partial_\mu K^0)(\partial^\mu \bar{K^0})\big) \\
& +\frac{d_2}{2f_K^2}\Bigg[\Big(\bar{p}p + \frac{5}{6}\bar{\Lambda^0}\Lambda^0 + \frac{1}{2}\bar{\Sigma^0}\Sigma^0 + \bar{\Sigma^+}\Sigma^+ + \bar{\Xi^-}\Xi^- + \bar{\Xi^0}\Xi^0\Big)(\partial_\mu K^+)(\partial^\mu K^-) \\
& +\Big(\bar{n}n + \frac{5}{6}\bar{\Lambda^0}\Lambda^0 + \frac{1}{2}\bar{\Sigma^0}\Sigma^0 + \bar{\Sigma^-}\Sigma^- + \bar{\Xi^-}\Xi^- + \bar{\Xi^0}\Xi^0\Big)(\partial_\mu K^0)(\partial^\mu \bar{K^0})\Bigg]. \qquad (2)
\end{aligned}
$$

The interaction Lagrangian modifying the energies of the $K(\bar{K})$-mesons is given as

see eq. (2) *on top of the page*

In (2) the first term is the vectorial interaction term (Weinberg-Tomozawa term) obtained from the kinetic term of the Lagrangian, the second and third terms are obtained from the explicit symmetry breaking and the pseudoscalar kinetic terms of the chiral effective Lagrangian, respectively [28,29]. The fourth and fifth terms in (2) for the KB interactions arise from the terms

$$\mathcal{L}^{BM}_{(d_1)} = \frac{d_1}{2}\,\mathrm{Tr}(u_\mu u^\mu)\,\mathrm{Tr}(\bar{B}B), \qquad (3)$$

and,

$$\mathcal{L}^{BM}_{(d_2)} = d_2\,\mathrm{Tr}(\bar{B}u_\mu u^\mu B). \qquad (4)$$

in the $SU(3)$ chiral model [27,28]. The last three terms in (2) represent the range term in the chiral model, with the last term being an isospin asymmetric interaction. We might note here that in eq. (2), we have not written the terms which are of the form $\bar{B}_i B_j$, with $i \neq j$. These types of terms are, however, relevant for the calculation of the kaon-nucleon scattering terms [29]. The Fourier transformation of the equation of motion for kaons (antikaons) leads to the dispersion relations

$$-\omega^2 + \vec{k}^2 + m_K^2 - \Pi(\omega, |\vec{k}|) = 0,$$

where Π denotes the kaon (antikaon) self-energy in the medium.

Explicitly, the self-energy $\Pi(\omega, |\vec{k}|)$ for the kaon doublet, (K^+, K^0) arising from the interaction (2) is given as

$$
\begin{aligned}
\Pi(\omega, |\vec{k}|) = & -\frac{1}{4f_K^2}\Big[3(\rho_p+\rho_n) \pm (\rho_p-\rho_n) \pm 2(\rho_{\Sigma^+}-\rho_{\Sigma^-}) \\
& -\big(3(\rho_{\Xi^-}+\rho_{\Xi^0}) \pm (\rho_{\Xi^-}-\rho_{\Xi^0})\big)\Big]\omega \\
& +\frac{m_K^2}{2f_K}(\sigma'+\sqrt{2}\zeta' \pm \delta') \\
& +\Big[-\frac{1}{f_K}(\sigma'+\sqrt{2}\zeta' \pm \delta') + \frac{d_1}{2f_K^2}(\rho_s^p + \rho_s^n \\
& +\rho^s{}_{\Lambda^0} + \rho^s{}_{\Sigma^+} + \rho^s{}_{\Sigma^0} + \rho^s{}_{\Sigma^-} + \rho^s{}_{\Xi^-} + \rho^s{}_{\Xi^0}) \\
& +\frac{d_2}{4f_K^2}\Big((\rho^s{}_p + \rho^s{}_n) \pm (\rho^s{}_p - \rho^s{}_n) + \rho^s{}_{\Sigma^0} \\
& +\frac{5}{3}\rho^s{}_{\Lambda^0} + (\rho^s{}_{\Sigma^+} + \rho^s{}_{\Sigma^-}) \pm (\rho^s{}_{\Sigma^+} - \rho^s{}_{\Sigma^-}) \\
& +2\rho^s{}_{\Xi^-} + 2\rho^s{}_{\Xi^0}\Big)\Big](\omega^2 - \vec{k}^2), \qquad (5)
\end{aligned}
$$

where the $\pm$ signs refer to the K^+ and K^0, respectively. In the above, $\sigma'(=\sigma-\sigma_0)$, $\zeta'(=\zeta-\zeta_0)$ and $\delta'(=\delta-\delta_0)$ are the fluctuations of the scalar-isoscalar fields σ and ζ, and the third component of the scalar-isovector field, δ, from their vacuum expectation values. The vacuum expectation value of δ is zero ($\delta_0 = 0$), since a nonzero value for it will break the isospin symmetry of the vacuum (the small isospin-breaking effect coming from the mass and charge difference of the up and down quarks has been neglected here). ρ_i and $\rho^s{}_i$ with $i = p, n, \Lambda, \Sigma^\pm, \Sigma^0, \Xi^-, \Xi^0$ are the number density and the scalar density of the baryon of type i.

Similarly, for the antikaon doublet, $(K^-, \bar{K^0})$, the self-energy is calculated as

$$\begin{aligned}\Pi(\omega, |\vec{k}\,|) = {} & \frac{1}{4f_K^2}\Big[3(\rho_p+\rho_n) \pm (\rho_p-\rho_n) \pm 2(\rho_{\Sigma^+}-\rho_{\Sigma^-}) \\ & -\big(3(\rho_{\Xi^-}+\rho_{\Xi^0}) \pm (\rho_{\Xi^-}-\rho_{\Xi^0})\big)\Big]\omega \\ & +\frac{m_K^2}{2f_K}(\sigma'+\sqrt{2}\zeta' \pm \delta') \\ & +\Big[-\frac{1}{f_K}(\sigma'+\sqrt{2}\zeta' \pm \delta') + \frac{d_1}{2f_K^2}(\rho_s^p+\rho_s^n \\ & +\rho^s{}_{\Lambda^0}+\rho^s{}_{\Sigma^+}+\rho^s{}_{\Sigma^0}+\rho^s{}_{\Sigma^-}+\rho^s{}_{\Xi^-}+\rho^s{}_{\Xi^0}) \\ & +\frac{d_2}{4f_K^2}\Big((\rho^s{}_p+\rho^s{}_n) \pm (\rho^s{}_p-\rho^s{}_n)+\rho^s{}_{\Sigma^0} \\ & +\frac{5}{3}\rho^s{}_{\Lambda^0}+(\rho^s{}_{\Sigma^+}+\rho^s{}_{\Sigma^-}) \pm (\rho^s{}_{\Sigma^+}-\rho^s{}_{\Sigma^-}) \\ & +2\rho^s{}_{\Xi^-}+2\rho^s{}_{\Xi^0}\Big)\Big](\omega^2-\vec{k}^2), \end{aligned} \tag{6}$$

where the $\pm$ signs refer to the K^- and $\bar{K^0}$, respectively. In eqs. (5) and (6), the number density and the scalar density of the baryon of type i are given by

$$\begin{aligned}\rho_i &= \frac{2}{(2\pi)^3}\int_0^{k_{Fi}} \mathrm{d}^3k, \\ \rho_i^s &= \frac{2}{(2\pi)^3}\int_0^{k_{Fi}} \mathrm{d}^3k \frac{M_i{}^*}{(k_{Fi}{}^2+M_i{}^{*2})^{1/2}}\,. \end{aligned} \tag{7}$$

with the baryon density, $\rho_B = \sum_i \rho_i$ and the baryon scalar density, $\rho_B{}^s = \sum_i \rho_i{}^s$. For low densities, $k_{Fi} \ll M_i$ and the scalar density has a value close to the number density of baryon i. However, as expected, there is appreciable deviation of $\rho_i{}^s$ from ρ_i, and hence in the value of $\rho_B{}^s$ as compared to ρ_B, at higher densities. For example, for $\eta = 0$ and $f_s = 0$, for the baryon density ρ_B (in fm^{-3}) as 0.075, 0.15, 0.3 and 0.6, the baryon scalar density $\rho_B{}^s$ (in fm^{-3}) has a value of 0.073, 0.142, 0.26 and 0.418, respectively. For the isospin asymmetric hadronic matter, these values for the scalar density are modified to 0.073, 0.141, 0.255 and 0.407 for $\eta = 0.3$, $f_s = 0$. There are modifications due to the presence of the hyperons in the medium. As it may be noted the scalar density has appreciable difference from the baryon density at higher values of ρ_B.

The optical potentials are calculated from the energies of the kaons and antikaons using

$$U(\omega, k) = \omega(k) - \sqrt{k^2+m_K^2}\,, \tag{8}$$

where m_K is the vacuum mass for the kaon (antikaon).

The parameters d_1 and d_2 are determined by a fit of the empirical values of the KN scattering lengths for $I = 0$ and $I = 1$ channels [28], which are taken as [30,41,42]

$$a_{KN}(I=0) \approx -0.09\,\mathrm{fm}, \qquad a_{KN}(I=1) \approx -0.31\,\mathrm{fm} \tag{9}$$

leading to the isospin-averaged KN scattering length as

$$\bar{a}_{KN} = \frac{1}{4}a_{KN}(I=0) + \frac{3}{4}a_{KN}(I=1) \approx -0.255\,\mathrm{fm}. \tag{10}$$

4 Results and discussions

The present calculations use the following model parameters. The values, $g_{\sigma N} = 10.6$, and $g_{\zeta N} = -0.47$ are determined by fitting vacuum baryon masses. The other parameters as fitted to the asymmetric nuclear-matter saturation properties in the mean-field approximation are: $g_{\omega N} = 13.3$, $g_{\rho N} = 5.5$, $g_4 = 79.7$, $g_{N\delta} = 2.5$, $m_\zeta = 1024.5\,\mathrm{MeV}$, $m_\sigma = 466.5\,\mathrm{MeV}$ and $m_\delta = 899.5\,\mathrm{MeV}$ [29]. The coefficients d_1 and d_2, calculated from the empirical values of the KN scattering lengths for $I = 0$ and $I = 1$ channels (9), are $2.56/m_K$ and $0.73/m_K$, respectively. However, one sees that there is appreciable dependence of the KN scattering lengths on the values of d_1 and d_2. For values of (d_1, d_2) as $(5.12/m_K, 0.73/m_K)$ and $(1.28/m_K, 0.73/m_K)$, one finds the values for the scattering lengths in the $I = 0$ and $I = 1$ channels, $(a_{KN}(I=0), a_{KN}(I=1))$ in fm as $(0.35, 0.13)$ and $(-0.31, -0.53)$, respectively, which are seen to be very different from the empirical values given by eq. (9). Similarly, to examine the sensitivity to the d_2-term, we note that for values of (d_1, d_2) as $(2.56/m_K, 1.46/m_K)$ and $(2.56/m_K, 0.36/m_K)$, one finds the scattering lengths in the $I = 0$ and $I = 1$ channels, $(a_{KN}(I=0), a_{KN}(I=1))$ in fm as $(-0.2158, -0.184)$ and $(-0.027, -0.373)$, respectively. We thus notice that the low-energy KN scattering lengths are quite sensitive to the parameters d_1 and d_2.

The kaon and antikaon properties were studied in isospin symmetric nuclear matter within the chiral $SU(3)$ model in ref. [27] and in the isospin asymmetric nuclear matter in refs. [28,29]. In the present work, we concentrate on the effects of the hyperons on the energies of the kaons and antikaons in strange hadronic matter. The contribution from the vector interaction (Weinberg-Tomozawa term) leads to a drop for the antikaon energy, whereas they are repulsive for the kaons in the asymmetric nuclear medium. There are now contributions from hyperons due to this vectorial interaction as given in the first term in eq. (2). One may note that there is no contribution arising from either Λ^0- or the Σ^0-hyperon, for such a baryon-pesudoscalar meson vectorial interaction. The scalar meson exchange term arising from the scalar-isoscalar fields (σ and ζ) is attractive for both K and $\bar{K}$ doublets. The first term of the range term of eq. (2) is repulsive, whereas the second term has an isospin symmetric attractive contribution for both kaons and antikaons, with contributions arising due to interactions with the baryon octet. The third term of the range term has an isospin dependence [29] which now takes into account the effects of hyperons, in addition to contributions from the nucleons.

The isospin asymmetries within the kaon doublet (K^+, K^0) as well as in antikaon $(K^-, \bar{K}^0)$ energies arise from the Weinberg-Tomazawa term due to asymmetry in proton and neutron number densities as well as asymmetries in the hyperon number densities. They also arise from the scalar-isovector δ field becoming nonzero for the isospin asymmetric hadronic matter as can be seen from the second term (the scalar exchange term) as well as the range term given by the third term in eq. (2). The d_2-term in the interaction Lagrangian given by eq. (2) also intro-

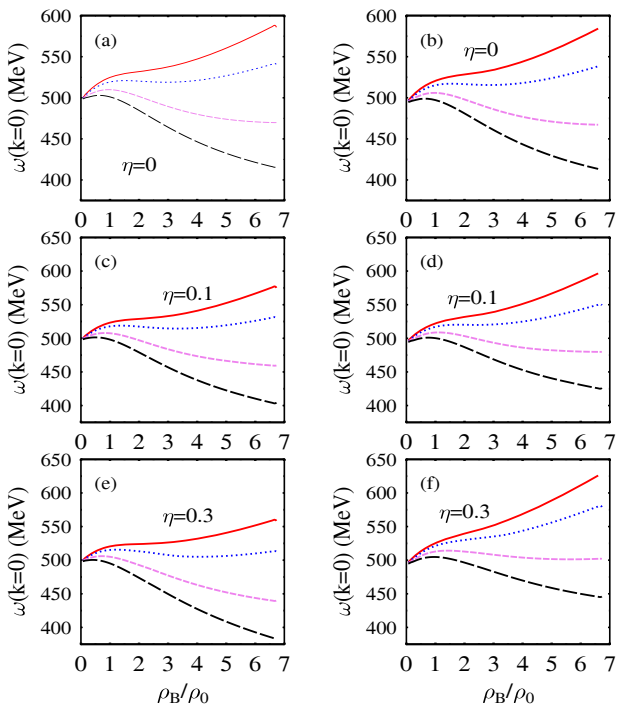

Fig. 1. The kaon energies (for K^+ in (a), (c), (e) and for K^0 in (b), (d) and (f)) at zero momentum, plotted as functions of the baryon density in units of nuclear-matter saturation density, ρ_B/ρ_0, for the isospin asymmetry parameter η as 0, 0.1 and 0.3, respectively. For each value of η, the energies are shown for $f_s = 0, 0.1, 0.3$ and 0.5, as the solid, dotted, short-dashed and long-dashed lines, respectively.

duces asymmetries for K^+ and K^0, as well as for antikaon (K^- and $\bar{K^0}$) energies, arising from both the nucleon as well as hyperonic sectors. For $\rho_n > \rho_p$, in the kaon sector, K^+ (K^0) has negative (positive) contributions from δ. The δ contribution from the scalar exchange term is positive (negative) for K^+ (K^0), whereas that arising from the range term has the opposite sign and dominates over the former contribution.

Figure 1 shows the variation of energy of kaons (K^+ and K^0) at zero momentum, with density for different values of isospin asymmetric parameter, η. At each value of η, the energies are shown for strangeness fractions, $f_s =$ 0, 0.1, 0.3 and 0.5. The strangeness fraction is defined by $f_s = \frac{\sum_i s_i \rho_i}{\rho_B}$ where s_i is the number of strange quarks of baryon i. One sees that for $f_s = 0$, there is an increase of the kaon energies at zero momentum, with an increase in density. However, one notices that as the strangeness fraction is increased to $f_s = 0.1$, for $\eta = 0$, there is a drop of the kaon energies after the initial rise, and then it is seen to increase again at still higher values of the density. When the strangeness fraction, f_s, is further increased ($f_s = 0.3$ and 0.5), one sees that the energies of the kaons decrease with increasing density. For $\eta = 0$, the decrease in energy of K^+ at zero momentum, as we go from $f_s = 0$ to $f_s =$ 0.5 is 23.65 MeV and 101 MeV for densities ρ_B as ρ_0 and $4\rho_0$, respectively. For K^0 the drop in the energy at zero momentum is 23.59 MeV at $\rho_B = \rho_0$ and 100.37 MeV at $\rho_B = 4\rho_0$, for $\eta = 0$.

At $\eta = 0.1$, $\rho_B = \rho_0$, the decrease in energy of K^+ is 25 MeV as we move from $f_s = 0$ to $f_s = 0.5$ and at $\rho_B = 4\rho_0$, it becomes 103.11 MeV. For K^0 the decrease is 22 MeV at $\rho_B = \rho_0$ and 98 MeV at $\rho_B = 4\rho_0$ for $f_s = 0.5$ as compared to the $f_s = 0$ case. For a higher isospin asymmetry ($\eta = 0.3$), the drop in the energy of K^+ is 25.34 MeV for $\rho_B = \rho_0$ and 104.4 MeV at $\rho_B = 4\rho_0$ for $f_s = 0.5$ as compared to the nonstrange hadronic matter. For K^0, the drop in the energies are given as 21 MeV at $\rho_B = \rho_0$ and 98 MeV at $\rho_B = 4\rho_0$. It is thus seen that for a given value of η, the energy of kaons decreases as we move to higher strangeness fractions. The decrease in energy is more pronounced at high densities. This is due to the fact that the second term of the range term (the d_1-term) which is attractive, becomes more and more dominant with increasing contributions from the hyperons, in addition to nucleons.

Also it is noted that for a given value of the strangeness fraction, the energy of K^+ decreases as we move to a higher value of the isospin asymmetry η and that of K^0 increases. At $f_s = 0$, $\rho_B = \rho_0$, the drop in the energy of K^+ is 4 MeV as we move from $\eta = 0.0$ to $\eta = 0.3$, whereas the energy of K^0 increases by 5 MeV for these conditions. The reason for the opposite behavior for the K^+ and K^0 on the isospin asymmetry in the nuclear medium [28,29] originates from the vectorial (Weinberg-Tomozawa), δ-meson cotribution as well as from the isospin-dependent range term (d_2-term) contributions. For K^+, the η-dependence of the energy is seen to be less senstive at higher densities, whereas the energy of K^0 is seen to have a larger drop from the $\eta = 0$ case, as we increase the density. The isospin asymmetry in the medium modification of the K^+- and K^0-mesons are seen to be particularly significant for higher strangeness fraction at high densities. For $f_s = 0.5$, at $\rho_B = \rho_0$, the drop (increase) in the energy of the K^+ (K^0) meson at zero momentum, as we move from $\eta = 0$ to $\eta = 0.3$, is 6 MeV (8 MeV), whereas these become about 19 MeV (26 MeV) at a density of $\rho_B = 4\rho_0$.

In fig. 2, we have shown the effects of the strangeness fraction on the energy of antikaons (K^- and $\bar{K^0}$) at zero momentum, with density, for given values of η. At given values of η, the energy of antikaons is found to increase as we move to higher strangeness fractions. For $\eta = 0$, $\rho_B = \rho_0$ the increase in the energy of K^- is 11.64 MeV and that of $\bar{K^0}$ is 11.63 MeV as we move from $f_s = 0$ to $f_s = 0.5$. At $\rho_B = 4\rho_0$, the energy of K^- and $\bar{K^0}$ are seen to increase by 23.47 MeV and 23.29 MeV, respectively. For $\eta = 0.3$, at $\rho_B = \rho_0$ the increase in the energy of K^- and $\bar{K^0}$ is 10 MeV and 14 MeV, respectively, as we move from $f_s = 0$ to $f_s = 0.5$. At $\rho_B = 4\rho_0$, the energy of K^- increases by 23.48 MeV and that of $\bar{K^0}$ increases by 21 MeV.

For the antikaons, the K^- ($\bar{K^0}$) energy at zero momentum is seen to increase (drop) with the isospin asymmetry parameter, η. The sensitivity of the isospin asymmetry dependence of the energies is seen to be larger for K^- with density as compared to $\bar{K^0}$. The isospin

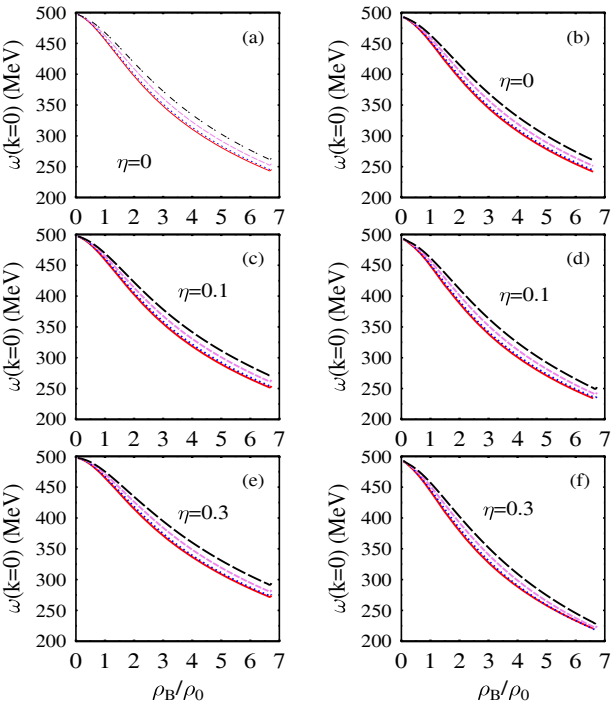

Fig. 2. The antikaon energies (for K^- in (a), (c), (e) and for $\bar{K}^0$ in (b), (d) and (f)) at zero momentum, plotted as functions of the baryon density in units of nuclear-matter saturation density, ρ_B/ρ_0, for the isospin asymmetry parameter η as 0, 0.1 and 0.3, respectively. For each value of η, the energies are shown for $f_s = 0, 0.1, 0.3$ and 0.5, as the solid, dotted, short-dashed and long-dashed lines, respectively.

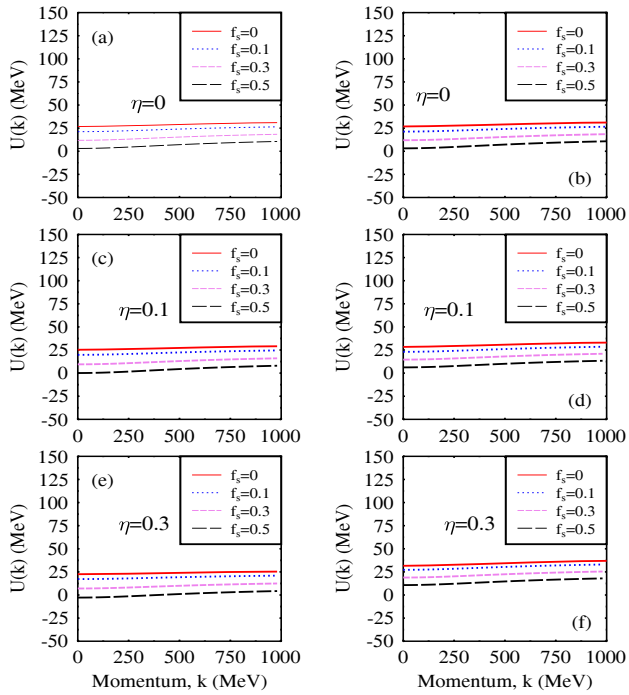

Fig. 3. The optical potentials for kaons (for K^+ in (a), (c) and (e) and for K^0 in (b), (d) and (f)) in MeV plotted as functions of the momentum for $\rho_B = \rho_0$ for $\eta = 0, 0.1$ and 0.3. For a given value of η, these are shown for $f_s = 0, 0.1, 0.3$ and 0.5, as the solid, dotted, short-dashed and long-dashed lines, respectively.

asymmetry gives rise to an increase in the mass of the K^--mesons as compared to the isopsin symmetric situation of $\eta = 0$, and hence would delay the onset on the K^- condensation to higher densities inside the neutron star matter. The changes in the mass of the K^- and $\bar{K}^0$ for $f_s = 0$, at $\eta = 0.3$ from the isospin symmetric case, are 9 MeV (8.61 MeV) at $\rho_B = \rho_0$, and at $\rho_B = 4\rho_0$ they are 25.86 MeV (19 MeV). For $f_s = 0.5$, these values are relatively weakly modified to 8 MeV (6 MeV) at $\rho_B = \rho_0$ and 26 MeV (22 MeV) for $\rho_B = 4\rho_0$.

The qualitative behavior of the strangeness dependences of the energies of the kaons and antikaons at finite momenta is reflected in their optical potentials plotted in figs. 3, 5 and 7 for the kaons, and in figs. 4, 6 and 8 for the antikaons, at selected densities, for the isospin asymmetry parameter, $\eta = 0$, 0.1 and 0.3, respectively. The different behavior of the K^+ and K^0, as well as for the K^- and $\bar{K}^0$ optical potentials in the dense asymmetric nuclear matter should be observed in their production as well as propagation in isospin asymmetric heavy-ion collisions. In particular, an experimental study of the K^+/K^0 as well as $K^-/\bar{K}^0$ ratios as well as their collective flow in the asymmetric heavy-ion collision experiments might be promising tools to investigate the isospin effects discussed here, whereas the kaon to antikaon ratios should show a significant change depending on the strangeness content of the matter. The effects of the isospin asymmetric

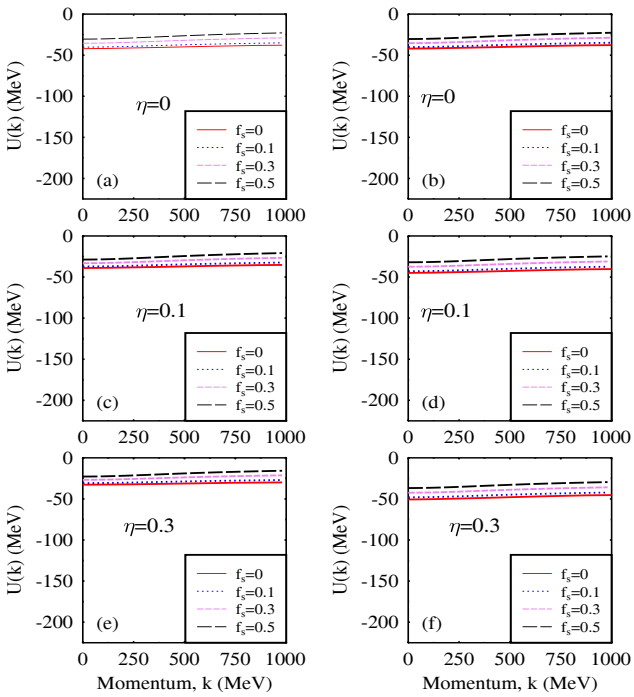

Fig. 4. The optical potentials for antikaons (for K^- in (a), (c) and (e) and for $\bar{K}^0$ in (b), (d) and (f)) in MeV plotted as functions of the momentum for $\rho_B = \rho_0$ for $\eta = 0, 0.1$ and 0.3. For a given value of η, these are shown for $f_s = 0, 0.1, 0.3$ and 0.5, as the solid, dotted, short-dashed and long-dashed lines, respectively.

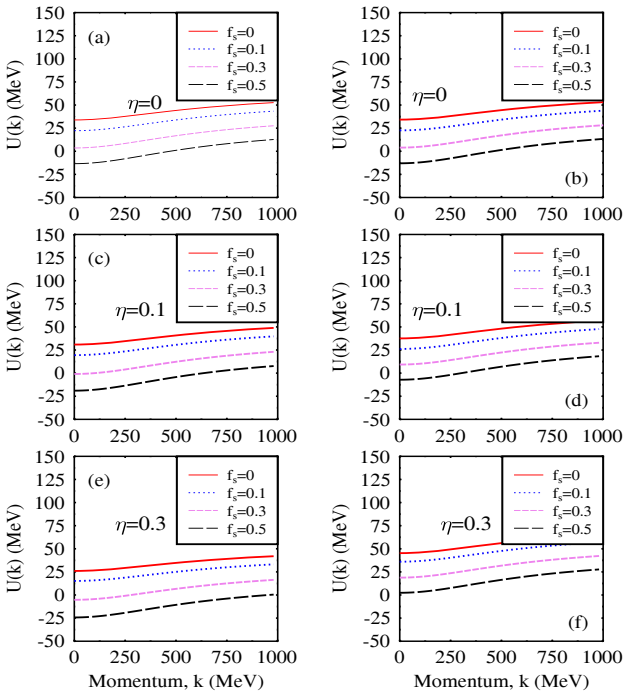

Fig. 5. The optical potentials for kaons (for K^+ in (a), (c) and (e) and for K^0 in (b), (d) and (f)) in MeV plotted as functions of the momentum for $\rho_B = 2\rho_0$ for $\eta = 0, 0.1$ and 0.3. For a given value of η, these are shown for $f_s = 0, 0.1, 0.3$ and 0.5, as the solid, dotted, short-dashed and long-dashed lines, respectively.

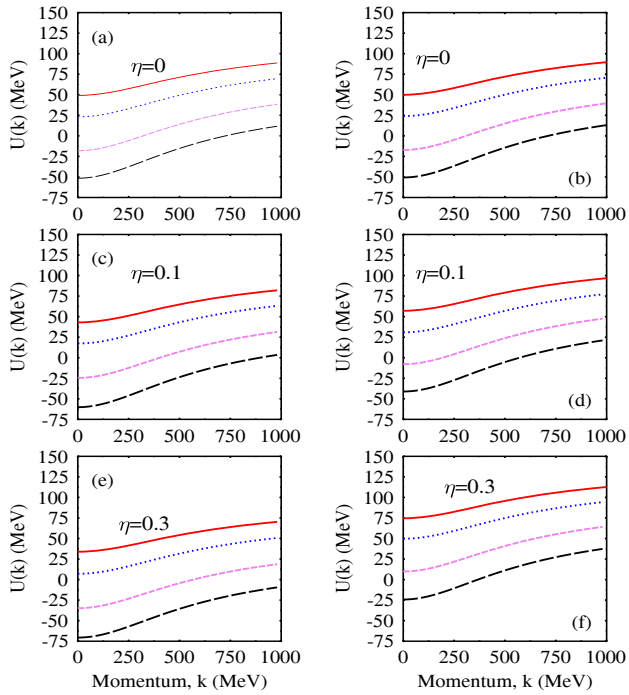

Fig. 7. The optical potentials for kaons (for K^+ in (a), (c) and (e) and for K^0 in (b), (d) and (f)) in MeV plotted as functions of the momentum for $\rho_B = 4\rho_0$ for $\eta = 0, 0.1$ and 0.3. For a given value of η, these are shown for $f_s = 0, 0.1, 0.3$ and 0.5, as the solid, dotted, short-dashed and long-dashed lines, respectively.

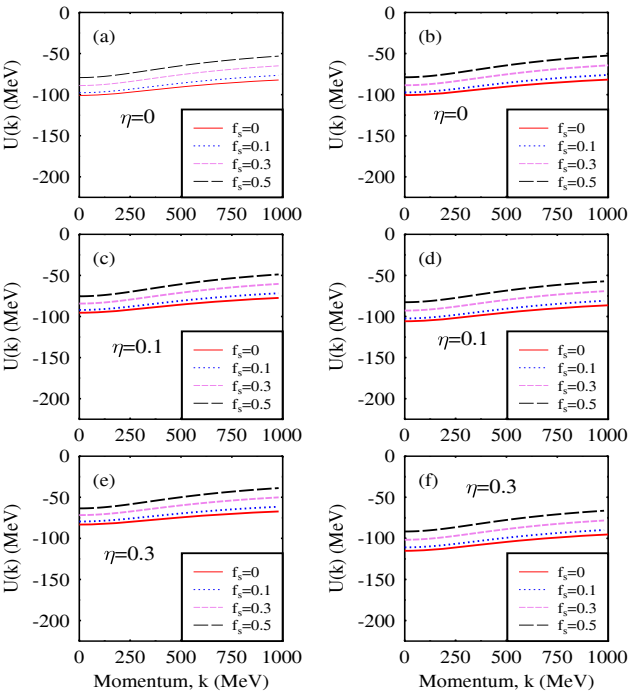

Fig. 6. The optical potentials for antikaons (for K^- in (a), (c) and (e) and for $\bar{K}^0$ in (b), (d) and (f)) in MeV plotted as functions of the momentum for $\rho_B = 2\rho_0$ for $\eta = 0, 0.1$ and 0.3. For a given value of η, these are shown for $f_s = 0, 0.1, 0.3$ and 0.5, as the solid, dotted, short-dashed and long-dashed lines, respectively.

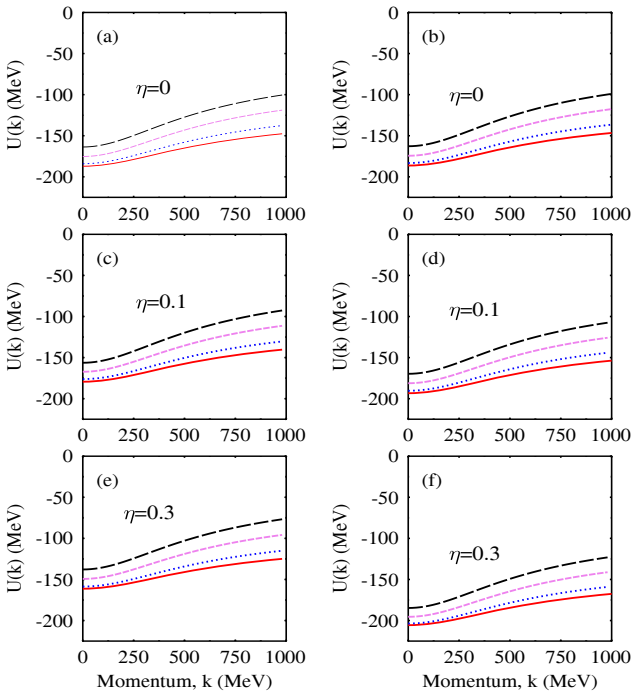

Fig. 8. The optical potentials for antikaons (for K^- in (a), (c) and (e) and for $\bar{K}^0$ in (b), (d) and (f)) in MeV plotted as functions of the momentum for $\rho_B = 4\rho_0$ for $\eta = 0, 0.1$ and 0.3. For a given value of η, these are shown for $f_s = 0, 0.1, 0.3$ and 0.5, as the solid, dotted, short-dashed and long-dashed lines, respectively.

and strangeness-dependent optical potentials could thus be observed in nuclear collisions at the CBM experiment at the proposed project FAIR at GSI, where experiments with neutron-rich beams are planned to be implemented. One might note here that the initial isospin asymmetry in heavy-ion collision experiments so far, corresponds to typical values of η not more than 0.15. However, with the radioactive ion beams to be used in CBM experiment at FAIR, GSI, one probably might access higher range of values for η and could observe effects arising from isospin asymmetry in the hot and dense hadronic matter.

5 Summary

To summarize, within a chiral $SU(3)$ model we have investigated the density dependence of the kaon and antikaon optical potentials in asymmetric hyperonic matter, arising from the interactions with the baryon octet (originating from a vectorial Weinberg-Tomozawa interaction, an isospin symmetric range term and an isospin asymmetric range term) and due to scalar-isoscalar mesons (σ, ζ) and scalar-isovector δ-mesons. The properties of the nucleons, hyperons and scalar mesons are modified in the hadronic medium and hence, due to their interactions with the kaons and antikaons, modifiy the $K(\bar{K})$-meson properties. The model with parameters fitted to reproduce the properties of hadron masses in vacuum, nuclear-matter saturation properties and low-energy KN scattering data, takes into account all terms up to the next-to-leading order arising in the chiral perturbative expansion for the interactions of $K(\bar{K})$-mesons with baryons.

There is a significant density dependence of the isospin asymmetry on the optical potentials of the kaons and antikaons. This dependence seems to be even more dominant for larger values of the strangeness fractions in dense hadronic matter. The results can be used in heavy-ion simulations that include mean fields for the propagation of mesons [39]. The different potentials of kaons and antikaons can be particularly relevant for neutron-rich heavy-ion beams at the CBM experiment at the future project FAIR at GSI, Germany, as well as at the experiments at the proposed Rare Isotope Accelerator (RIA) laboratory, USA. The K^+/K^0 and $K^-/\bar{K}^0$ ratios as well as their flow pattern for different isospin of projectile and target are promising observables to study these effects. In contrast, the kaon to antikaon ratios are especially sensitive to the strangeness content of the matter as the optical potential of the kaons, as compared to the antikaons, is strongly dependent on the amount of strangeness in the system. Furthermore, the medium modification of antikaons due to isospin asymmetry in dense matter can have important consequences, for example on the onset of antikaon condensation in the bulk charge neutral matter in neutron stars. The effects of finite temperatures on optical potentials of kaons and antikaons and their possible implications on high-energy heavy-ion collision experiments are under investigation.

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